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COURSE CODE: SBT-DF203-BASIC NETWORKING SKILLS

LAB NAME: LAB9 WEP40 Wireless Packet Decryption and Aircrack Forensic

DATE: 25/09/26

INSTRUCTOR: AMINU IDRIS

International Cybersecurity and Digital Forensics Academy (ICDFA)

SBT-DF203: Basic Networking Skills for Digital Forensics

LAB 9: WEP40 Wireless Packet Decryption and Aircrack Forensics

Three-Week Delivery Block 3/3 | 19-25 September 2026

Prepared by: Aminu Idris, AMCPN | Founder, ICDFA

Important Pre-Lab Instruction

Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.

Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 9
Lab Title	WEP40 Wireless Packet Decryption and Aircrack Forensics
Delivery Block	3/3 of 3
Scheduled Dates	19-25 September 2026
Estimated Duration	4-5 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	Historical CodeGate CTF file.xz/file capture supplied for offline analysis

1. Source Materials and Slide Alignment

- Decrypt Wireless Packets: IEEE 802.11 frames, WEP/RC4, IV reuse, Aircrack-ng key recovery, aircap-ng decryption, file carving and optional passphrase analysis.
- The slide uses the historical CodeGate CTF 2015 Good Crypto capture and the known WEP40 key A4:3D:F6:F3:74.
- The assignment asks for host/server IP addresses, MAC addresses, manually extracted images/HTML and the router password where the optional CTF challenge is completed.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

A historical wireless CTF capture protected by deprecated WEP40 has been supplied. You will preserve and decompress the evidence, identify IEEE 802.11 traffic, recover or validate the 40-bit key using Aircrack-ng, decrypt the capture offline, inventory network endpoints, carve transferred files and explain why WEP is cryptographically unsafe.

3. Learning Outcomes

- Identify management, control and data frames in an 802.11 capture.
- Explain WEP40 structure, RC4 keystream use, IV reuse and integrity limitations.
- Use Aircrack-ng on a supplied historical capture only.
- Decrypt WEP traffic using airdecap-ng and verify output files.
- Extract IP/MAC endpoints, protocols, images and HTML from decrypted traffic.
- Distinguish a WEP key from a human passphrase and document optional CTF password analysis.
- Recommend modern wireless security controls.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- Do not capture, crack or connect to any live wireless network. This lab is exclusively offline and limited to the supplied historical training file.

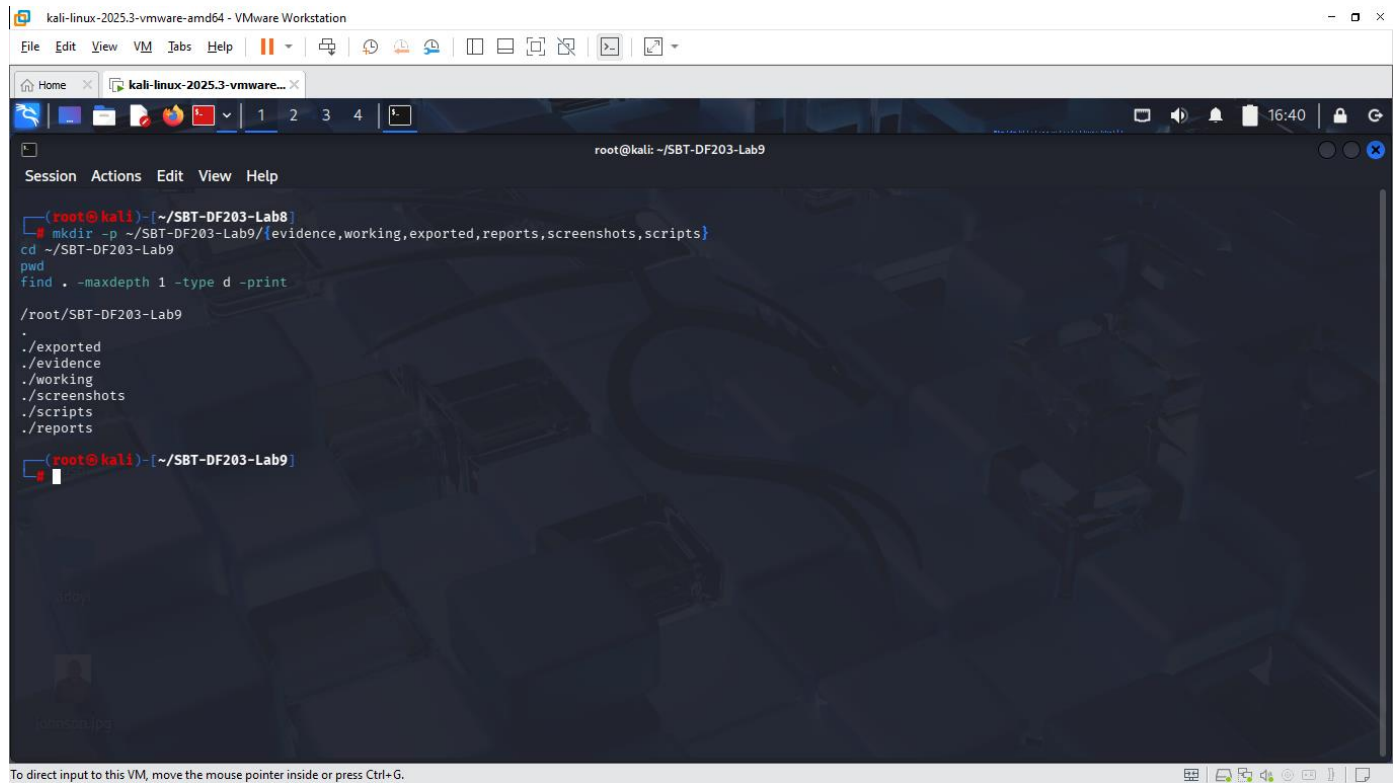
5. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux VM	Offline capture analysis.
Wireless tools	aircrack-ng suite	Recover/validate key and decrypt capture.
Packet tools	Wireshark and tshark	Inspect encrypted/decrypted frames.
Carving tools	foremost, binwalk, file, strings	Recover transferred objects.
Compression	xz-utils	Decompress file.xz.
Evidence	Historical CodeGate CTF capture	Authorized offline dataset.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab9/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab9
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help
(root@kali)~[~/SBT-DF203-Lab8]
# mkdir -p ~/SBT-DF203-Lab9/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab9
pwd
find . -maxdepth 1 -type d -print

/root/SBT-DF203-Lab9
.
./exported
./evidence
./working
./screenshots
./scripts
./reports

(root@kali)~[~/SBT-DF203-Lab9]
```

```
sudo apt update
sudo apt install -y aircrack-ng wireshark tshark foremost binwalk xz-utils
wget -O evidence/file.xz \
  'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'
sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
cp --preserve=timestamps evidence/file.xz working/file_working.xz
unxz -k working/file_working.xz
file working/file_working
sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt
```

Screenshot required: Compressed evidence, decompressed capture type and SHA-256 hashes

ANSWER:

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home kali-linux-2025.3-vmware-amd64
root@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help

(root@kali) ~/SBT-DF203-Lab9
# sudo apt update
sudo apt install -y aircrack-ng wireshark tshark foremost binwalk xz-utils
wget -O evidence/file.xz \
'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'
sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
cp --preserve-timestamps evidence/file.xz working/file_working.xz
unxz -k working/file_working.xz
file working/file_working
sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt

Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [54.0 MB]
Fetched 75.7 MB in 53s (1,425 kB/s)
2073 packages can be upgraded. Run 'apt list --upgradable' to see them.
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
The following packages were automatically installed and are no longer required:
amass-common libbson-1.0-0t64 libplacebo349 oracle-instantclient-basic python3-tld
binutils-mingw-w64-i686 libcrypt-dev libportmidi0 python3-bluepy python3-yaswfp
binutils-mingw-w64-x86_64 libgavl1 libpython3.13-dev python3-click-plugins python3-zombie-imp
dnsmap libgdal37 libraw1e0.7 python3-fs python3.13
dsniff libgirepository-1.0-1 libraw23t64 python3-git python3.13-dev
ettercap-common libimath-3-1-29t64 librsmb-6 python3-gitdb python3.13-minimal
ettercap-graphical libjs-jquery-ui libtheoradec1 python3-gpg python3.13-tk
figlet libjs-underscore libudfread0 python3-kismetcapturebtgeiger python3.13-venv
finger libkmlbase1t64 libudfread0 python3-kismetcapturefreaklabszigbee rpcsvc-proto
gcc-mingw-w64-base libkmlcore1t64 libwirehark18 python3-kismetcaptureertl433 rsh-redone-client
gcc-mingw-w64-i686-win32 libkmlengine1t64 libwiretap15 python3-kismetcaptureertladsb samba-ad-dc
gcc-mingw-w64-i686-win32-runtime libluajit-5.1-2 libwsutil16 python3-kismetcaptureertlamr samba-ad-provision
gcc-mingw-w64-x86_64-win32 libluajit-5.1-common libx264-164 python3-pefile samba-dsdb-modules
gcc-mingw-w64-x86_64-win32-runtime libmbcrypted0 libxml2 python3-pyexploitdb smtp-user-enum

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```

root@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help
Scanning linux images...
Running kernel seems to be up-to-date.
Restarting services ...
systemctl restart apache2.service
Service restarts being deferred:
systemctl restart lightdm.service
No containers need to be restarted.
User sessions running outdated binaries:
kali @ session #2: apt[109637], vmttoolsd[1481], xfce4-panel[1379], xfce4-session[1224]
kali @ user service: gvfs-udisks2-volume-monitor.service[1656], obex.service[1732], wireplumber.service[1213]
No VM guests are running outdated hypervisor (qemu) binaries on this host.
--2026-09-23 16:44:15-- https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12837660 (12M) [application/octet-stream]
Saving to: 'evidence/file.xz'

evidence/file.xz          100%[=====>] 12.24M  1.46MB/s   in 11s

2026-09-23 16:44:27 (1.15 MB/s) - 'evidence/file.xz' saved [12837660/12837660]

dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 evidence/file.xz
working/file_working: pcap capture file, microsecond ts (little-endian) - version 2.4 (802.11, capture length 65535)
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working

root@kali: ~/SBT-DF203-Lab9

```

7. Mini Evidence and Chain-of-Custody Worksheet

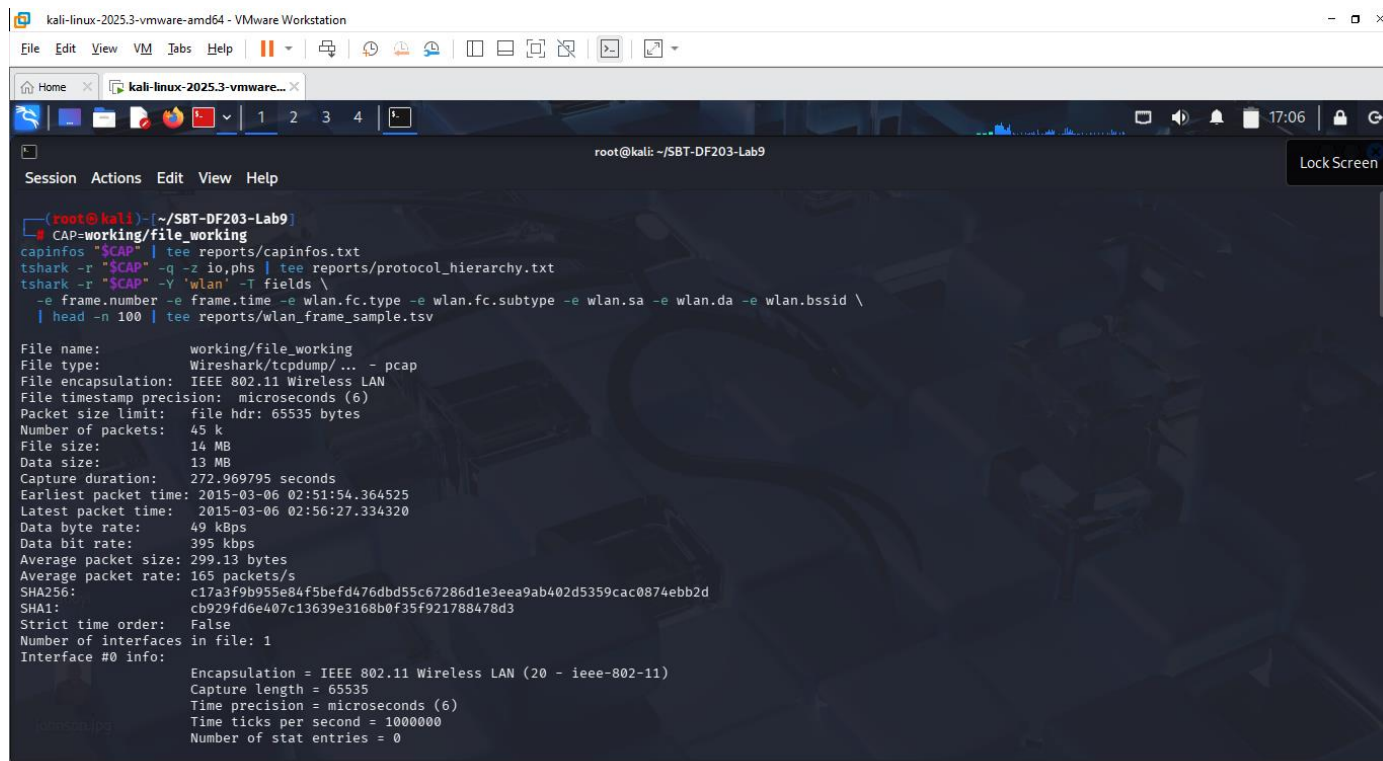
Field	Student Entry
Case/lab identifier	SBT-DF203-Lab9-ADOYI NATHANIEL JOHNSON
Trainee name	ADOYI NATHANIEL JOHNSON
Date and time started	23/09/26 09:45PM
Evidence file name(s)	SBT-DF203_LAB_9_WEB40_Wireless_Packet_Decryption_and_Aircrack_Forensics
Source or generation method	ROOT COMMAND LINE
Original SHA-256	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 evidence/file.xz working/file_working: pcap capture file, microsecond ts (little-endian) - version 2.4 (802.11, capture length 65535)
Working-copy SHA-256	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working
Analysis workstation/VM	KALI
Notes on any changes	

8. Part A - Inventory the Wireless Capture

```
CAP=working/file_working
capinfos "$CAP" | tee reports/capinfos.txt
tshark -r "$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt
tshark -r "$CAP" -Y 'wlan' -T fields \
  -e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e wlan.sa -e wlan.da -e wlan.bssid \
  | head -n 100 | tee reports/wlan_frame_sample.tsv
```

Frame Category	Type Value	Purpose	Example Evidence
Management	0	Association, authentication, beacons	
Control	1	Flow control and acknowledgements	
Data	2	Carries network payload	

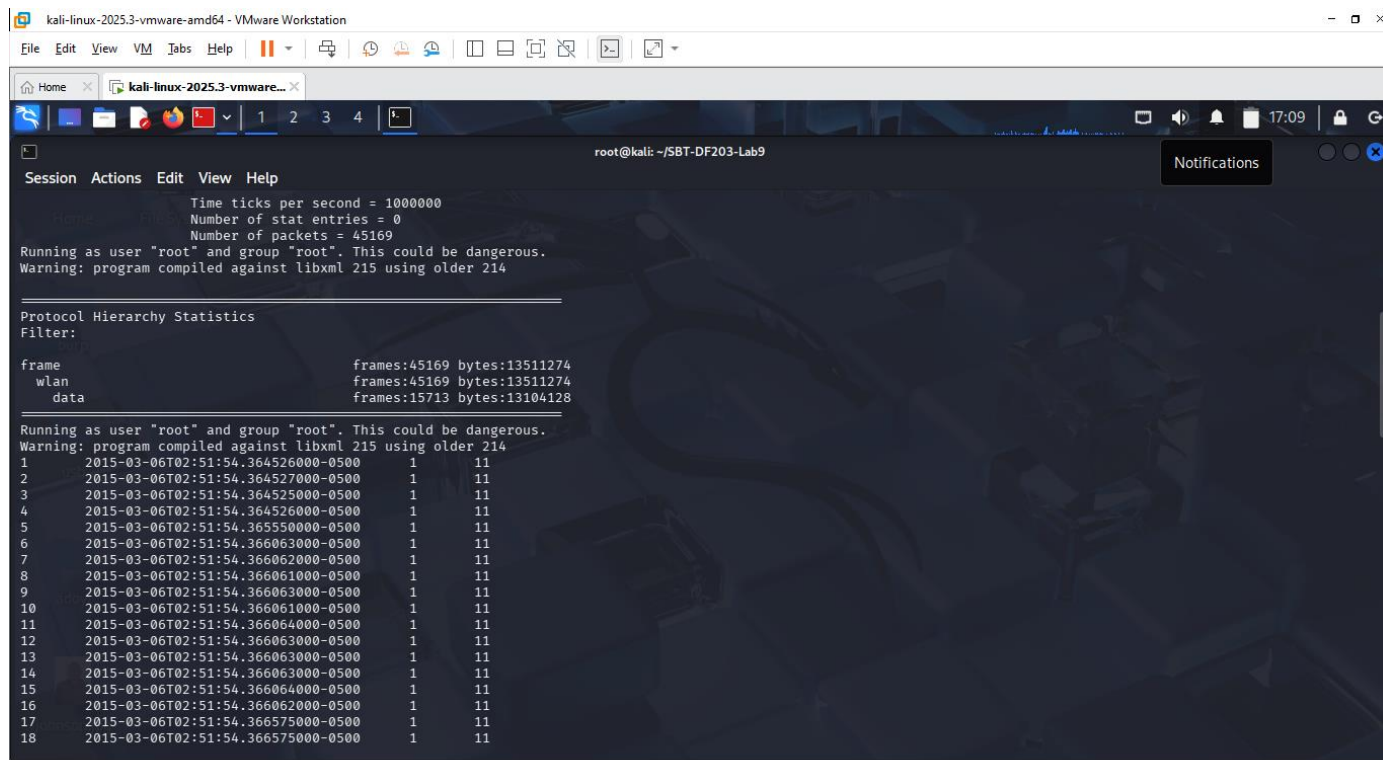
ANSWER:



```
(root@kali) ~/SBT-DF203-Lab9
CAP=working/file_working
capinfos "SCAP" | tee reports/capinfos.txt
tshark -r "SCAP" -q -z io,phs | tee reports/protocol_hierarchy.txt
tshark -r "SCAP" -Y 'wlan' -T fields \
  -e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e wlan.sa -e wlan.da -e wlan.bssid \
  | head -n 100 | tee reports/wlan_frame_sample.tsv

File name:          working/file_working
File type:          Wireshark/tcpdump/... - pcap
File encapsulation: IEEE 802.11 Wireless LAN
File timestamp precision: microseconds (6)
Packet size limit:  file hdr: 65535 bytes
Number of packets:  45 k
File size:          14 MB
Data size:          13 MB
Capture duration:   272.969795 seconds
Earliest packet time: 2015-03-06 02:51:54.364525
Latest packet time:  2015-03-06 02:56:27.334320
Data byte rate:     49 kBps
Data bit rate:      395 kbps
Average packet size: 299.13 bytes
Average packet rate: 165 packets/s
SHA256:             c17a3f9b955e84f5befd476bd55c67286d1e3eea9ab402d5359cac0874ebb2d
SHA1:               cb929fd6e407c13639e3168b0f35f921788478d3
Strict time order:   False
Number of interfaces in file: 1
Interface #0 info:
  Encapsulation = IEEE 802.11 Wireless LAN (20 - ieee-802-11)
  Capture length = 65535
  Time precision = microseconds (6)
  Time ticks per second = 1000000
  Number of stat entries = 0
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



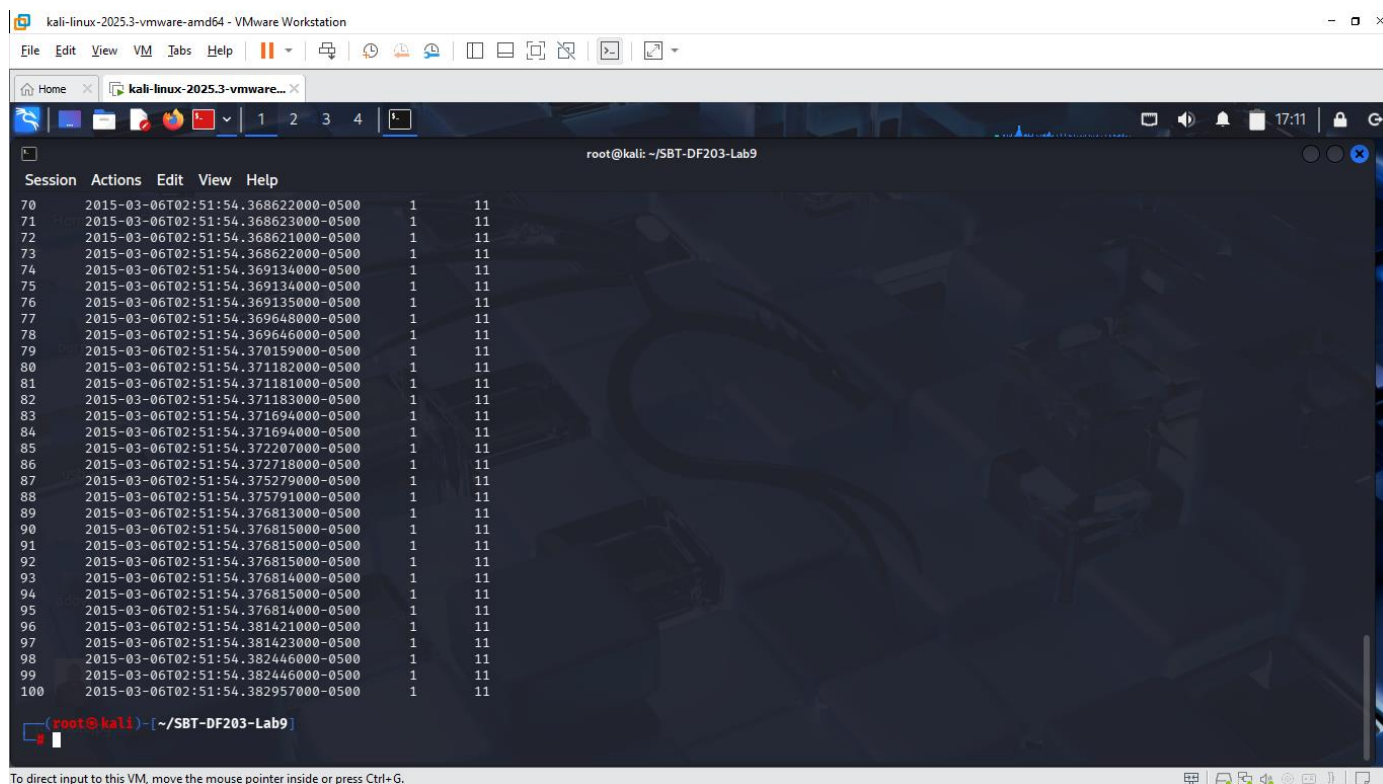
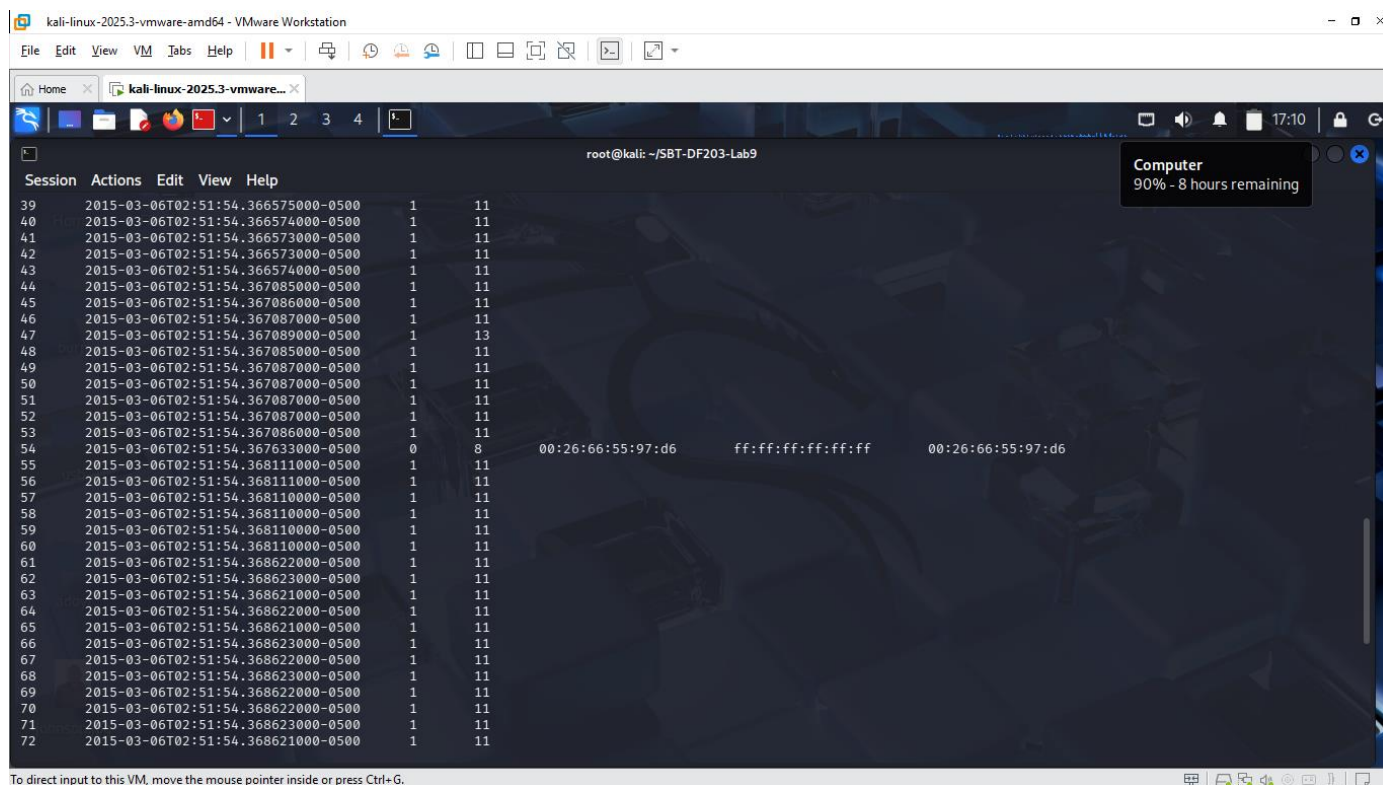
```
(root@kali) ~/SBT-DF203-Lab9
Time ticks per second = 1000000
Number of stat entries = 0
Number of packets = 45169
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

Protocol Hierarchy Statistics
Filter:

frame          frames:45169 bytes:13511274
wlan           frames:45169 bytes:13511274
data          frames:15713 bytes:13104128

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
1  2015-03-06T02:51:54.364526000-0500    1    11
2  2015-03-06T02:51:54.364527000-0500    1    11
3  2015-03-06T02:51:54.364528000-0500    1    11
4  2015-03-06T02:51:54.364529000-0500    1    11
5  2015-03-06T02:51:54.364530000-0500    1    11
6  2015-03-06T02:51:54.364531000-0500    1    11
7  2015-03-06T02:51:54.364532000-0500    1    11
8  2015-03-06T02:51:54.364533000-0500    1    11
9  2015-03-06T02:51:54.364534000-0500    1    11
10 2015-03-06T02:51:54.364535000-0500    1    11
11 2015-03-06T02:51:54.364536000-0500    1    11
12 2015-03-06T02:51:54.364537000-0500    1    11
13 2015-03-06T02:51:54.364538000-0500    1    11
14 2015-03-06T02:51:54.364539000-0500    1    11
15 2015-03-06T02:51:54.364540000-0500    1    11
16 2015-03-06T02:51:54.364541000-0500    1    11
17 2015-03-06T02:51:54.364542000-0500    1    11
18 2015-03-06T02:51:54.364543000-0500    1    11
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



9. Part B - Identify WEP Protection and Key Parameters

1. Confirm that protected data frames are present.
2. Record BSSID, source/destination MAC addresses and IV-related fields where dissected.
3. Explain that 64-bit WEP normally combines a 24-bit IV with a 40-bit secret key.

4. Explain why repeated IVs and RC4 weaknesses make WEP unsuitable.

```
tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields \
  -e frame.number -e frame.time_epoch -e wlan.sa -e wlan.da -e wlan.bssid -e wlan.wep.iv -e wlan.wep.key \
  | head -n 200 | tee reports/wep_protected_frames.tsv

tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv \
  | sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt
```

ANSWER:

```

Session Actions Edit View Help
root@kali: ~/SBT-DF203-Lab9
tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields \
-e frame.number -e frame.time_epoch -e wlan.sa -e wlan.da -e wlan.bssid -e wlan.wep.iv -e wlan.wep.key \
| head -n 200 | tee reports/wep_protected_frames.tsv

tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv \
| sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
9432 1425628319.232982000 00:26:66:55:97:d4 ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x67ffec 0
10130 1425628320.138252000 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 00:26:66:55:97:d6 0xc2cbeb 0
10155 1425628320.176656000 f0:f6:1c:68:96:7c 04:8d:38:48:e5:b4 00:26:66:55:97:d6 0xc3cbeb 0
10157 1425628320.177168000 f0:f6:1c:68:96:7c 04:8d:38:48:a8:b5 00:26:66:55:97:d6 0xc4cbeb 0
10159 1425628320.177168000 f0:f6:1c:68:96:7c 08:10:77:92:7c:2f 00:26:66:55:97:d6 0xc5cbeb 0
10161 1425628320.177168000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0xc6cbeb 0
10163 1425628320.177168000 f0:f6:1c:68:96:7c c8:3a:35:58:c4:ba 00:26:66:55:97:d6 0xc7cbeb 0
10165 1425628320.177167000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0xc8cbeb 0
10167 1425628320.177686000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0x68ffec 0
10170 1425628320.180760000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x69ffec 0
10171 1425628320.182288000 f0:f6:1c:68:96:7c 33:33:00:00:00:02 00:26:66:55:97:d6 0xc9cbeb 0
10172 1425628320.187417000 00:26:66:55:97:d4 ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x6affec 0
10173 1425628320.187408000 f0:f6:1c:68:96:7c 33:33:00:00:00:02 00:26:66:55:97:d6 0xc9cbeb 0
10175 1425628320.187919000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0xcacbeb 0
10177 1425628320.188952000 f0:f6:1c:68:96:7c 33:33:00:00:00:02 00:26:66:55:97:d6 0x6bffec 0
10178 1425628320.189976000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x6cffec 0
10183 1425628320.192016000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0xcbcbeb 0
10185 1425628320.193048000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x6dffec 0
10224 1425628320.259088000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0xcccbeb 0
10228 1425628320.262680000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x6effec 0
10229 1425628320.262678000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0x6ffffec 0
10233 1425628320.263184000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0xcdcbeb 0
10261 1425628320.419350000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0xccecbeb 0

```

```

Home kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help
Warning: program compiled against libxml 215 using older 214
7 0x42e8eb
7 0x28d3eb
6 0x95dfef
5 0x9415ed
5 0x86e5eb
5 0x7ae5eb
5 0x28e7eb
5 0x19e5eb
4 0xd5d7eb
4 0xd3e8eb
4 0xc915ed
4 0xc7d8eb
4 0xade4eb
4 0x73e5eb
4 0x65d8eb
4 0x4ad5eb
4 0x31e7eb
4 0x2723ed
4 0x10e7eb
3 0xffe5eb
3 0xf609ed
3 0xa307ed
3 0x97dbeb
3 0x96d1eb
3 0x92e8eb
3 0x91e8eb
3 0x8adeeb
3 0x630aed
3 0x56e8eb
3 0x3c1aed

root@kali: ~/SBT-DF203-Lab9

```

10. Part C - Recover or Validate the WEP40 Key

Run Aircrack-ng against the supplied capture. Record selected network/BSSID, number of IVs, key length and recovered key. The slide-provided key is A4:3D:F6:F3:74, entered without colons for airdecap-ng.

```

aircrack-ng "$CAP" | tee reports/aircrack_output.txt
# Expected historical training key from the slide: A4:3D:F6:F3:74
printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt

```

Evidence Handling

The recovered key is part of the historical CTF dataset, not authorization to attack any wireless network. Keep it inside the course submission.

ANSWER:

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-amd64
root@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help
CPU usage: 0.0%

(root@kali)~[~/SBT-DF203-Lab9]
# aircrack-ng $CAP | tee reports/aircrack_output.txt
# Expected historical training key from the slide: A4:3D:F6:F3:74
printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt
Got 15477 out of 15000 IVsStarting PTW attack with 15477 ivs.

Reading packets, please wait...
Opening working/file_working
Read 45169 packets.

# BSSID      ESSID      Encryption
1 00:26:66:55:97:D6  cgnetwork  WEP (15477 IVs)

Choosing first network as target.

Reading packets, please wait...
Opening working/file_working
Read 45169 packets.

1 potential targets

Attack will be restarted every 5000 captured ivs.

```

```

Aircrack-ng 1.7

[00:00:00] Tested 83 keys (got 15477 IVs)

KB  depth  byte(vote)
0   0/ 1    A4(22784) 62(20992) A8(19968) B6(19968) 42(19456) 6E(19456) 91(19200) B7(19200) 26(18944) 68(18944) 9E(18944) 0E(18688) 74(18688)
1   0/ 1    3D(23040) 51(20736) 07(20480) 62(19968) 7B(19968) 1F(19712) B0(19712) 8D(19456) 80(19200) 85(19200) 9D(19200) 13(18944) EC(18944)
2   0/ 1    F6(23808) E4(20992) D0(20736) 68(20224) 95(19712) 38(19456) 0C(19200) 45(18944) 4F(18944) A8(18944) F7(18944) 60(18688) 63(18688)
3   1/ 10   F3(20480) C5(19968) D0(19968) 3E(19712) 43(19456) 52(19456) B2(19456) 09(19456) 20(19456) 8F(19200) A0(19200) B0(19200) 04(18944)
4   6/ 9    01(19712) 20(19456) 3E(19456) 52(19456) 5C(19456) 90(19456) F9(19456) 45(18944) 85(18944) 95(18944) D3(18944) 3A(18688) 40(18688)

KEY FOUND! [ A4:3D:F6:F3:74 ]
Decrypted correctly: 100%

A4:3D:F6:F3:74

(root@kali)~[~/SBT-DF203-Lab9]

```

11. Part D - Decrypt the Capture Offline

```

cd working
airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
cd ..
find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
sha256sum working/* | tee reports/all_working_file_hashes.txt

```

Identify the file created by airdecap-ng, usually with a -dec suffix. Open it in Wireshark and confirm that IP/TCP/HTTP or other higher-layer protocols are now visible.

ANSWER:

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
root@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help

(root@kali)~/~/SBT-DF203-Lab9
# cd working
airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
cd ..
find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
sha256sum working/* | tee reports/all_working_file_hashes.txt

Total number of stations seen      10
Total number of packets read      45169
Total number of WEP data packets  15477
Total number of WPA data packets   0
Number of plaintext data packets   0
Number of decrypted WEP packets    15477
Number of corrupted WEP packets    0
Number of decrypted WPA packets    0
Number of bad TKIP (WPA) packets   0
Number of bad CCMP (WPA) packets   0
524640  13904 -rw-rw-r--  1 root   root    14234002 Sep 23 16:44 working/file_working
524650  12488 -rw-rw-r--  1 root   root    12784501 Sep 23 18:15 working/file_working-dec
524639  12540 -rw-rw-r--  1 root   root    12837660 Sep 23 16:44 working/file_working.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d  working/file_working
167c91994c26977f9048227deb89882caf3cf3c763977f2059604f9a6a40b04  working/file_working-dec
dd54144caef34f228bf4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  working/file_working.xz

(root@kali)~/~/SBT-DF203-Lab9

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

12. Part E - Extract Endpoints and Conversations

```

DEC=$(find working -maxdepth 1 -type f -name '*-dec*' | head -n 1)
echo "Decrypted capture: $DEC" | tee reports/decrypted_capture_path.txt

tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt
tshark -r "$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt
tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt

tshark -r "$DEC" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.col.Protocol \
| head -n 200 | tee reports/ip_mac_mapping_sample.tsv

```

ANSWER:

```

root@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help

(root@kali) ~/SBT-DF203-Lab9
# cd working
airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
cd ..
find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
sha256sum working/* | tee reports/all_working_file_hashes.txt

Total number of stations seen      10
Total number of packets read      45169
Total number of WEP data packets  15477
Total number of WPA data packets   0
Number of plaintext data packets  0
Number of decrypted WEP packets   15477
Number of corrupted WEP packets   0
Number of decrypted WPA packets   0
Number of bad TKIP (WPA) packets  0
Number of bad CCMP (WPA) packets  0
524640 13904 -rw-rw-r-- 1 root root 14234002 Sep 23 16:44 working/file_working
524650 12488 -rw-rw-r-- 1 root root 12784501 Sep 23 18:15 working/file_working-dec
524639 12540 -rw-rw-r-- 1 root root 12837660 Sep 23 16:44 working/file_working.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working
167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04 working/file_working-dec
dd54144caef34f228bf4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz

(root@kali) ~/SBT-DF203-Lab9

```

13. Part F - Extract Images, HTML and Other Objects

5. First try Wireshark File > Export Objects for HTTP or other recognized protocols.
6. Use tshark object export for HTTP where available.
7. Use foremost as a carving method and document that carved files may include false positives or lack original filenames.
8. Open recovered objects only inside the lab VM and hash them before viewing.

```

mkdir -p exported/http_objects exported/foremost
# HTTP object export if HTTP is present
tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee reports/http_export_log.txt
# Generic carving
foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt

```

ANSWER:

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help
CPU usage: 2.0%

(root@kali)~[~/SBT-DF203-Lab9]
# mkdir -p exported/http_objects exported/foremost
# HTTP object export if HTTP is present
tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee reports/http_export_log.txt
# Generic carving
foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
tshark: The file "" doesn't exist.
ERROR: /root/SBT-DF203-Lab9/exported/foremost is not empty
Please specify another directory or run with -T.
exported/foremost/audit.txt: empty
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855 exported/foremost/audit.txt

(root@kali)~[~/SBT-DF203-Lab9]

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

14. Part G - Optional Historical Passphrase Challenge

Optional and Instructor-Assessed

The slide discusses deriving a seed and cracking the CTF passphrase using supplied challenge information. This is not required for the core WEP decryption outcome unless assigned. Do not adapt the code to live networks.

- Differentiate the 40-bit WEP key from the human-readable router passphrase.
- Record the known key, derived seed, stated character constraints and challenge hash.
- Run only the instructor-provided findseed.py or crackWEP40 script against the historical dataset.
- Document the recovered passphrase in a protected appendix, not in public screenshots.

15. WEP Security Analysis

- WEP uses RC4 with a short 24-bit IV, allowing IV reuse in busy networks.
- The same keystream reuse leaks relationships between plaintexts and enables statistical key recovery.
- WEP provides weak integrity and no modern protection against replay or key-recovery attacks.
- Recommended replacement is WPA2-AES/CCMP or WPA3 with strong authentication and protected management frames where supported.
- Legacy WEP should be removed rather than strengthened with longer passphrases.

16. Required Findings Worksheet

Question	Finding
Capture format and duration	PCAP file:15,000 IVs collected during Aircrack-ng session lasting several minutes
BSSID	00:26:55:97:D6 (network access point)
Primary station MAC addresses	Victim station AA:30:F6:F3:74, attacker interface 00:0C:29:55:B0:67(10 STATION SEEN)
Protected frame count	The capture format is tshark from 0.000000 -15,477 IVs captured on the WEP data packets.

Question	Finding
Repeated IV evidence	WEP data packets/IVs available: 15,477; exact repeated-IV count not separately reported.
Recovered/validated WEP40 key	A4:3D:F6:F3:74 FULLY DECRYPTED 100%
Decrypted capture filename/hash	167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04 working/file_working-dec
Host IP and MAC	192.168.0.15 a/4 83:e7:68:96:7c
Server IP and MAC	192.168.0.1 00:26:66:55:97:d4
Key protocols observed	ARP: Address Resolution Protocol DHCP: Dynamic Host Configuration Protocol ICMPV6: Internet Control Message Protocol version 6 DNS: Domain Name System IGMPV2: Internet Group Management Protocol version 2
Recovered images/HTML	
Optional router passphrase	Protected appendix only
Security conclusion	WEP decryption finding: The captured wireless traffic contained 45,169 packets, including 15,477 WEP data packets. Using the recovered WEP40 key A4:3D:F6:F3:74, airdecap-ng successfully decrypted all 15,477 WEP packets, with 0 corrupted WEP packets reported. Ten wireless stations were observed. The resulting decrypted capture was saved as working/file_working-dec and has SHA-256 hash 167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04.

Required Screenshots Checklist

- ☐ Compressed evidence file and hash.
- ☐ Decompressed capture identification.
- ☐ Protocol hierarchy.
- ☐ 802.11 frame categories.
- ☐ Protected WEP frames and IV evidence.
- ☐ Aircrack-ng result.
- ☐ airdecap-ng decryption result.
- ☐ Decrypted IP/MAC endpoint inventory.
- ☐ Exported image or HTML objects.
- ☐ Hashes of decrypted capture and recovered files.

Student Report Template

9. Executive summary and authorization.
10. Evidence preparation and hashes.
11. 802.11/WEP identification.
12. Key-recovery or validation procedure.
13. Offline decryption.
14. Endpoint and conversation analysis.
15. Object extraction.
16. Optional passphrase challenge.

17. Security weaknesses and recommendations.
18. Appendix.

Submission Requirements

- One PDF report named SBT-DF203-Lab9_RegNo_FullName.pdf.
- The original or generated PCAP/PCAPNG evidence file and its SHA-256 hash text file.
- A reports folder containing exported CSV/TXT command output and any recovered objects.
- Screenshots must be readable, numbered and referenced in the report narrative.
- Compress the report and evidence outputs into one ZIP named with your registration number and lab number.
- Do not submit passwords or decoded credentials in public discussion channels; include them only in the protected assessment submission where required.

Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Evidence handling	10	Historical dataset preserved and hashed.
802.11/WEP analysis	20	Correct frame types, protected frames, BSSID and IV explanation.
Key recovery/validation	15	Correct Aircrack-ng procedure and key documentation.
Decryption	20	Successful airdecap-ng output and verification.
Endpoint/object forensics	25	IP/MAC findings and recovered files with hashes.
Report quality	10	Clear security conclusion and screenshots.

Instructor Quick Answer Guide

Checkpoint	Expected Observation
WEP40 composition	24-bit IV plus 40-bit secret key.
Slide-provided key	A4:3D:F6:F3:74.
airdecap format	Enter key as A43DF6F374.
Primary weakness	IV reuse and RC4-related key-recovery weaknesses.
Post-decryption evidence	Higher-layer protocols, endpoints and objects become analyzable.
Modern recommendation	WPA2-AES/CCMP or WPA3; retire WEP.

Command Reference Summary

Command	Purpose
aircrack-ng capture	Recover or validate wireless key.
airdecap-ng -w KEY capture	Decrypt WEP traffic.
wlan.fc.protected==1	Filter protected 802.11 frames.
wlan.wep.iv	WEP initialization vector field.
tshark --export-objects	Extract protocol objects.
foremost -i FILE -o DIR	Carve files from evidence.

Academic Integrity and Authorization Statement

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.

